

HMS Elite

Hospital Management System
Technical Documentation & User Manual



Prepared for: Team Lead

Version: 1.0.0 | March 2026

HMS: Elite Hospital Management System

Technical Documentation & User Manual — v1.0.0

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1. Executive Summary

The HMS Elite (Hospital Management System) is a production-grade, full-stack application designed to streamline healthcare operations. Built with a React 19 frontend and a FastAPI (Python) backend secured with JWT authentication, it provides a seamless, data-persistent experience for hospital administrators, doctors, and clinical staff across 13 specialised modules.

2. Technology Stack

Component	Technology	Purpose
Frontend	React 19, Vite 7, Bootstrap 5	Responsive single-page application
Backend	FastAPI (Python), Uvicorn	High-performance async REST API
Database	SQLite 3, SQLAlchemy ORM	Persistent relational data storage
Authentication	JWT (HS256, python-jose)	Signed tokens, 24-hour expiry
Password Security	PBKDF2-HMAC-SHA256	Salted hash, 100,000 iterations
Styling	CSS Variables, Glassmorphism	Light/Dark theme support

3. System Architecture

The system follows a classic **client-server architecture** with a clear separation of concerns:

- Client Layer**: React SPA — handles routing, state management, and UI rendering. All API calls use `Authorization: Bearer` headers.
- API Layer**: FastAPI RESTful backend — validates JWT on every protected route, enforces per-user data isolation via `owner_user_id`.
- Data Layer**: SQLite managed via SQLAlchemy ORM — full CRUD for all 13 entities across 12 modules.

4. Security Model

Dimension	Implementation
Authentication	HS256 JWT issued at login/signup; validated on every API request
Authorization	Every record scoped to <code>owner_user_id</code> ; no cross-user data leakage
Password Storage	PBKDF2-HMAC-SHA256 with random 32-byte salt; timing-safe comparison
CORS	Configurable via <code>ALLOWED_ORIGINS</code> environment variable
Token Expiry	24 hours (configurable via <code>ACCESS_TOKEN_EXPIRE_MINUTES</code>)

5. How to Run the Project

Start Backend

```
cd HMS/backend npm run dev
```

```
``` cd HMS npm run dev ```
```

6. API Reference

Module	Endpoint Prefix	Operations
Authentication	`/api/auth`	POST /login, /signup
Patients	`/api/patients`	GET, POST, PUT, DELETE
Appointments	`/api/appointments`	GET, POST, PUT, DELETE
Dashboard	`/api/dashboard`	GET (Live Metrics)

7. Core Modules — Visual Gallery

Each module presents a dual interface: a high-level **Table View** for monitoring and a task-focused **Form View** for data entry and administrative actions.

7.1 Clinical Operations Dashboard

Real-time hospital metrics: active admissions, critical alerts, total patients, and monthly revenue.

**Table View:**

 **HMS Core**  
v1.0.0 Premium

Command Center

Patient Directory

Scheduling

Medical Records

Revenue Cycle

Pharmacy

Diagnostics & Lab

Facility Management

Human Capital

Intelligence

Hospital Logistics

Emergency Blood Bank

Doc Admin

Admin

CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Clinical Operations Center

Monitor hospital activity and manage patient workflows.

Export Report

+ Add Patient

00

ACTIVE ADMISSIONS

+Live

00

CRITICAL ALERTS

0 Labs

00

TOTAL PATIENTS

0 Staff

₹0

MONTHLY REVENUE

0 Bills

Department Performance

Last 7 Days

Telemetry Unavailable

Live clinical telemetry data will appear here once analytics sensors are synchronized.

Live Medical Queue

Queue Empty

No patient appointments are currently queued.

View All Appointments

7.2 Patient Registry

Full patient lifecycle management — registration through discharge. **Table View:**

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 **HMS Core**

v1.0.0 Premium

Command Center

**Patient Directory**

Scheduling

Medical Records

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

## Patient Registry

Manage patient records and hospital admissions.

Register Patient

Active Patients

Search IDs...

ID	PATIENT NAME	BLOOD GROUP	LAST VISIT	STATUS	ACTIONS
No Patients Registered  Your clinical registry is currently empty. Start by registering your first patient.  Register Patient					
Showing 0 active profiles					

Previous1Next

Form View (Registration):

Register New Patient

FULL NAME

e.g. John Doe...

AGE

0

GENDER

Male

BLOOD GROUP

O+

ADMISSION STATUS

Outpatient

Cancel

Register Patient

Form Details:

Field Label	Input Type	Description
Full Name	Text	Legal name of the patient
Age	Number	Current age in years
Gender	Select	Male, Female, Other
Blood Group	Select	A+, B+, O+, AB+, etc.
Admission Status	Select	Admitted, Outpatient, Discharged

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7.3 Appointment Scheduling

Table View:

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v1.0.0 Premium

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Admin

CORE OPERATIONAL  
System Health OPTIMAL

Search patients, records, modules...

Clinic Schedule

Manage patient appointments and clinical slots.

Today 0

Book Appointment

Daily Queue

Today 0Completed 0

SLOT TIME	PATIENT NAME	ATTENDING DOCTOR	VISIT TYPE	STATUS	ACTIONS
   No Appointments  The daily clinical schedule is clear. You can book a new appointment for any registered patient.  + Book Appointment					

Form View (Booking):

Schedule Appointment

SELECT PATIENT

Select a registered patient...

DOCTOR

Enter doctor name

VISIT TYPE

Checkup

TIME SLOT

14:30 PM

Cancel

Confirm Appointment

Form Details:

Field Label	Input Type	Description
Patient Name	Select	Registered patient reference
Department	Select	Hospital department
Doctor Name	Select	Assigned clinician
Date	Date	Scheduled date

## 7.4 Electronic Medical Records (EMR)

### Table View:

 **HMS Core**  
v1.0.0 Premium

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**Medical Records**

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Facility Management

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Admin

CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Electronic Medical Records

New EMR Entry

Encrypted Record History

Backend records loaded: 0

RECORD ID	CLINICAL ID	ENTRY DATE	PATIENT IDENTITY	CLINICIAN	PRIMARY DIAGNOSIS	ELECTRONIC VALIDATION
   No Clinical Records  Electronic health history is currently empty for the selected clinical scope.  New EMR Entry						

### Form View (New Entry):

Electronic Clinical Entry Protocol

PATIENT IDENTITY

Select an active patient profile...

CLINICAL DIAGNOSIS TELEMETRY

e.g. Acute Respiratory Infection

MEDICAL PRESCRIPTION PROTOCOL

List pharmaceutical interventions...

ATTENDING CLINICIAN

Enter clinician name

CLINICAL REFERENCE ID

CID-948431

Cancel

Finalize Entry

### Form Details:

Field Label	Input Type	Description
Patient ID	Select	Clinical record link
Diagnosis	Text	Findings
Prescription	Textarea	Medication list

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7.5 Billing & Finance

Table View:

 **HMS Core**  
v1.0.0 Premium

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Financial Records

Manage clinical billing and payment records.

Create New Invoice

Recent Invoices

Export CSV

Sync Ledger

INVOICE ID	PATIENT NAME	DATE	AMOUNT	STATUS	METHOD	ACTIONS
No Invoices  No financial records or billing cycles have been initialized.  + Create New Invoice						

Form View (Invoice):

Generate Invoice

PATIENT NAME

Select a patient...

AMOUNT (INR)

INR 0.00

Invoices are initialized as pending until payment is finalized.

Cancel

Generate Invoice

Form Details:

Field Label	Input Type	Description
Patient Name	Select	Billing recipient
Total Amount	Number	Total cost
Payment Method	Select	Cash, Card, Insurance

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### Table View:

### Form View (Stock):

### Form Details:

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7.7 Laboratory Management

Table View:

 **HMS Core**  
v1.0.0 Premium

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Diagnostic Laboratory

Order New Test

Active Diagnostic Queue

LAB ID	PATIENT IDENTITY	DIAGNOSTIC TEST	ORDERING CLINICIAN	PROTOCOL STATUS	ELECTRONIC VALIDATION
No Lab Tests  The diagnostic queue is currently empty. Place an order for pathology or radiology tests.  Order New Test					

Form View (Order):

Laboratory Test Ordering Protocol

SUBJECT IDENTITY

Select a registered patient profile...

DIAGNOSTIC TEST CLASSIFICATION

e.g. Complete Blood Count (CBC) / MRI Brain

ORDERING PHYSICIAN

Enter clinician name

Cancel

Finalize Order

Form Details:

Field Label	Input Type	Description
Patient ID	Select	Test subject
Test Name	Text	Procedure name

7.8 Bed Management

Table View:

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v1.0.0 Premium

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CORE OPERATIONAL

System HealthOPTIMAL

Search patients, records, modules...

Bed Management

Monitor real-time bed availability and patient admissions.

AVAILABLEOCCUPIED+ Add Bed

ICU0 TOTAL

No Beds Configured

This ward currently has no beds registered in the system.

+ Add Bed

General Ward0 TOTAL

No Beds Configured

**Form View (Add Bed):**

## 7.9 Human Resources

 **HMS Core**

v1.0.0 Premium

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Human Resources & Staffing

Add Staff Member

Active Personnel Directory

Departments 0 On Duty 0

STAFF MEMBER	ROLE CLASSIFICATION	CLINICAL DEPARTMENT	ACTIVE SHIFT	DUTY STATUS	ELECTRONIC VALIDATION
No Staff Onboarded  The personnel directory is empty. Add healthcare professionals to manage departments and shifts.  Add Staff Member					

Form View (Onboard):

Personnel Onboarding Protocol

FULL LEGAL NAME

Enter staff member name

ROLE CLASSIFICATION

Enter role

CLINICAL DEPARTMENT

Enter department

ASSIGNED SHIFT

Enter shift

Cancel

Finalize Onboarding

7.10 Reports & Analytics

Table View:

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Reports & Analytics

Hospital operational analytics generated from live backend data.

Download All

25 Mar 2026, 13:21

OPERATIONAL

OPD Daily Report (0 Appointments)

Last Generated: 25 Mar 2026, 13:21

RUN SYNC →

25 Mar 2026, 13:21

FINANCIAL

Pharmacy & Billing Summary (₹0)

Last Generated: 25 Mar 2026, 13:21

RUN SYNC →

25 Mar 2026, 13:21

INVENTORY

Inventory Stock Level (0 Alerts)

Last Generated: 25 Mar 2026, 13:21

RUN SYNC →

25 Mar 2026, 13:21

PERFORMANCE

Consultation Efficiency (0 Doctors Active)

Last Generated: 25 Mar 2026, 13:21

RUN SYNC →

GEN-AI STRATEGIC ADVISORY

Hospital data systems are currently in a fresh state. Historical and live analytics will populate this advisory once clinical operations commence.

Review Setup Guide

7.11 Inventory

Table View:

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v1.0.0 Premium

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CORE OPERATIONAL

System Health

OPTIMAL

Search patients, records, modules...

Inventory Management

Track and manage hospital supplies and equipment.

Audit Report

Add Supply

ITEM ID	ITEM NAME	CATEGORY	CURRENT STOCK	CONSUMPTION	STATUS	ACTION
No Inventory Items  Your hospital supply repository is currently empty. Start by adding new supplies or equipment.  Add Supply						

Form View (Supply):

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Add Inventory Item

×

ITEM NAME

CATEGORY

UNIT

INITIAL STOCK

Cancel

Add Item

## 7.12 Blood Bank

### Table View:

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v1.0.0 Premium

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CORE OPERATIONAL

System Health

OPTIMAL

Blood Bank Repository

Monitor real-time blood unit availability and donation cycle.

Request Registry

Add Donor

Inventory Empty

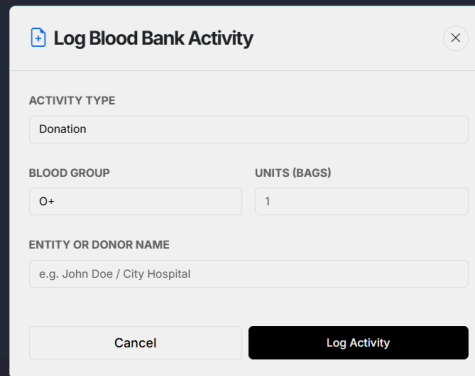
Blood bank units have not been initialized.

Recent Activity Logs

TYPE	GROUP	UNITS	ENTITY/PERSON	TIMESTAMP
No Activity   No donation or usage cycles have been logged yet.				

### Form View (Log):

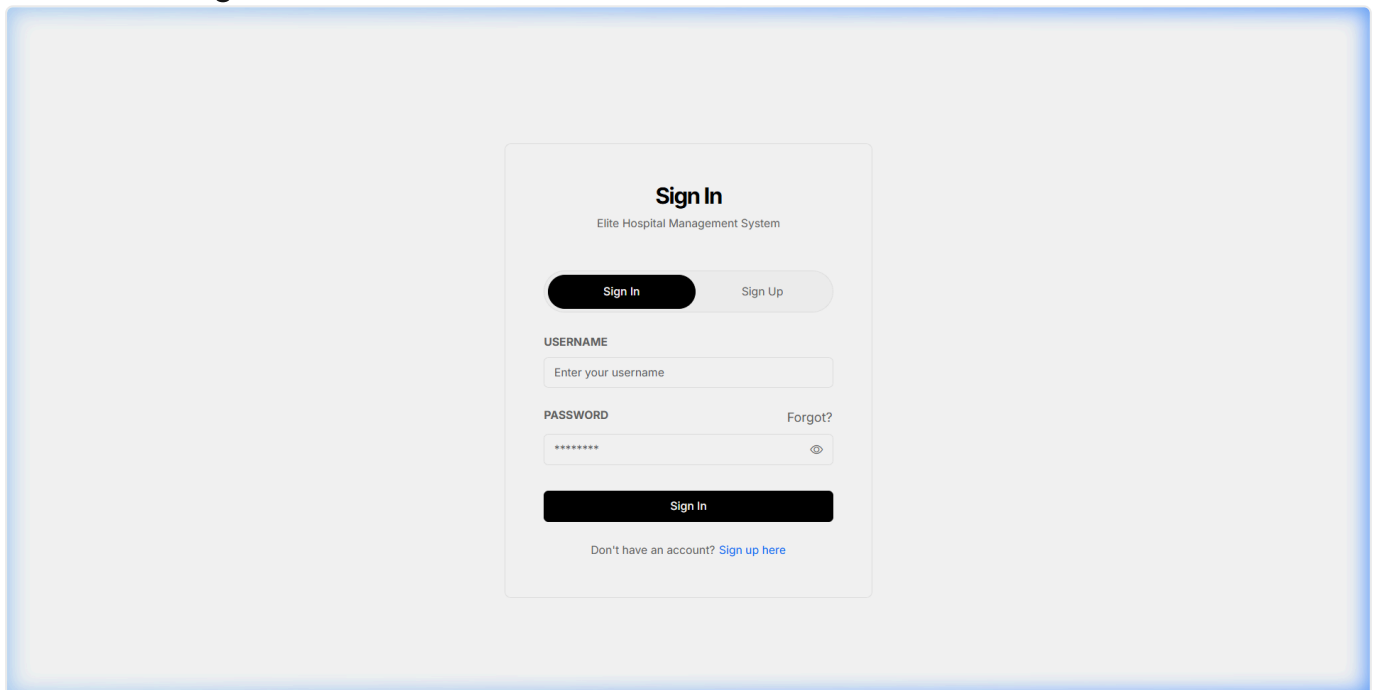
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A modal form titled "Log Blood Bank Activity" with a close button in the top right corner. The form contains the following fields: "ACTIVITY TYPE" with a dropdown menu showing "Donation"; "BLOOD GROUP" with a dropdown menu showing "O+"; "UNITS (BAGS)" with a text input field showing "1"; and "ENTITY OR DONOR NAME" with a text input field showing "e.g. John Doe / City Hospital". At the bottom, there are two buttons: "Cancel" and "Log Activity".

## 7.13 Authentication

### Form View (Login):



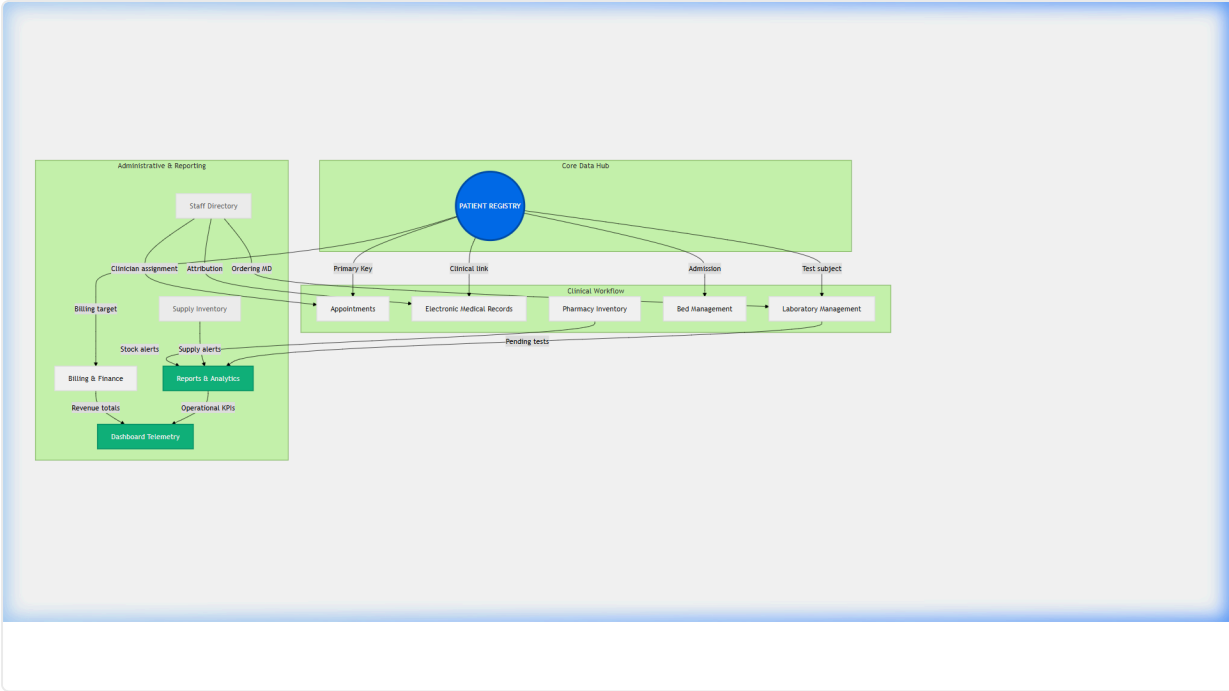
A "Sign In" form for the "Elite Hospital Management System". The form has a title "Sign In" and a subtitle "Elite Hospital Management System". Below the subtitle are two buttons: "Sign In" and "Sign Up". The form contains two input fields: "USERNAME" with a placeholder "Enter your username" and "PASSWORD" with a placeholder "\*\*\*\*\*" and a "Forgot?" link. Below the input fields is a "Sign In" button. At the bottom, there is a link "Don't have an account? Sign up here".

## 10. System Interconnectivity & Data Flow

The HMS Elite is a highly integrated ecosystem where data flows seamlessly between specialised modules to provide a 360-degree view of hospital operations.

### 10.1 Visual Architecture Map

The following diagram illustrates the primary data flows and relational links between the 13 core modules.



### 10.2 Primary Data Relationships

- **\*\*Patient ID\*\*** is the primary key linking Patients, Appointments, EMR, Billing, Lab, and Beds.
- **\*\*Staff (Doctors)\*\*** provide clinical attribution to Appointments, EMR, and Laboratory orders.
- **\*\*Financial Telemetry\*\*** from Billing flows into the Dashboard for real-time oversight.
- **\*\*Resource Alerts\*\*** from Pharmacy and Inventory flow into the Reports module.

### 10.3 Relational Mapping Table

Module A	Module B	Relationship Type	Data Flow
Patients	All Clinical Mod.	One-to-Many	Patient ID reference
Staff (Doctors)	Medical	Many-to-Many	Clinician assignment
Finance (Billing)	Dashboard	Financial	Revenue summation

**Note:** This documentation reflects HMS Elite v1.0.0 — production-ready with JWT authentication, full CRUD persistence, and comprehensive architectural mapping.